

Intern Roles & Responsibilities

1

Intern 1 — Backend & Auth

Node.js · PostgreSQL · Prisma · JWT · NextAuth

Auth System

Database

APIs

- Set up PostgreSQL + Prisma schema (all models)
- Build NextAuth.js for 5 roles: Super Admin, School Admin, Teacher, Student, Parent
- JWT + refresh token rotation, bcrypt password hashing
- RBAC middleware — strict role-based route guards on all endpoints
- School data isolation: every DB query filtered by schoolId
- Fee privacy API — school admin sees only their own school's fee
- Rate limiting + Zod input validation on all API routes
- Seed script: 3 demo schools, 50 students, 1 full course with 10 lessons
- Unit tests: auth middleware, fee isolation, lesson unlock logic

2

Intern 2 — Super Admin Panel

Next.js · Course Builder · FFmpeg/HLS · OpenAI · Recharts

Super Admin

Course Builder

Video Upload

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- Super Admin dashboard: schools, revenue, courses, analytics overview
- School management: add/edit/delete, unique ID + auto-password generation
- Per-school fee management — private and isolated per school
- Drag-and-drop course builder: Grade → Modules → Lessons → Assessments
- Video upload flow: FFmpeg transcoding → HLS → S3 → signed expiring URLs
- Toggle lesson lock/unlock, publish/unpublish courses, live preview
- AI quiz generator: OpenAI API → MCQ / drag-drop / match JSON output
- Announcements: push to specific school or all schools
- Analytics: enrollment, completion rates, scores, revenue charts

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Intern 3 — School Admin Panel

Next.js · Attendance · Fees · Library · Bus · PDF Reports

School Admin

Attendance

Fee Mgmt

- School Admin dashboard: student count, attendance %, fee collection status
- Student registration: bulk CSV import, auto roll numbers, ID card print
- Daily + period-wise attendance marking; monthly % auto-calculation
- Absentee alerts via email/SMS to parents
- Fee management: fee structure, payments, PDF receipt generation
- Library: book catalog, issue/return tracking, fine calculation
- Bus management: routes, stops, student allocation, Google Maps embed
- Teacher management: assign classes, subjects, communication tools
- Reports: PDF/Excel export for attendance, fees, library

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Intern 4 — Student Gamified Dashboard

Next.js · Framer Motion · Phaser.js · Lottie · PWA

Gamification

Video Player

Mini Games

- Gamified home: XP bar, level system, streak counter, Lottie mascot guide
- Sequential video player with unlock logic (pass quiz ≥60% to proceed)
- Quiz engine: MCQ, drag-drop, match — auto-triggered after every lesson
- World Map: modules = islands, lessons = levels; locked/foggy/done states
- Badges, class-level leaderboard (monthly reset), avatar shop with coins
- Phaser.js mini-games between lessons for engagement
- Cartoon UI for Grade 1–5: Nunito/Fredoka font, bright colors, large targets
- Mobile-first responsive design + PWA support for offline access
- Celebration animations: XP gain, badge unlock, level up sparkles

Project Timeline — 3 Phases

Phase 1 — MVP Core		Weeks 1–3
Intern 1	Prisma schema, auth system (all 5 roles), RBAC middleware, seed data	
Intern 2	Super Admin: school management UI + course builder (upload + lesson creation)	
Intern 3	School Admin: student registration + daily attendance marking	
Intern 4	Student: login screen, basic dashboard, sequential video player + quiz	
Phase 2 — Full Features		Weeks 4–7
Intern 1	Fee isolation API, signed video URLs, unit tests, rate limiting + Zod validation	
Intern 2	AI quiz generator, analytics dashboard, announcements, video HLS pipeline	
Intern 3	Fee management + receipt PDF, library, bus management, reports export	
Intern 4	World Map, badges, leaderboard, Phaser.js mini-games, Grade 1–5 cartoon UI	

Phase 3 — Polish & Launch

Weeks 8–10

■ Intern 1	Parent portal APIs, FCM notifications, Razorpay payment integration, Docker Compose
■ Intern 2	Advanced analytics, certificate PDF auto-generation, live preview in course builder
■ Intern 3	Teacher tools, communication system, attendance SMS/email alerts
■ Intern 4	PWA offline support, Lottie mascot polish, sound effects, avatar shop coins

■ Collaboration Rule:	Interns 2, 3, 4 consume Intern 1's APIs. Week 1 — use mock data. Week 2 onwards — swap to live APIs. Intern 1 must share API contracts (Postman/OpenAPI) by end of Week 1. Daily sync between Intern 1 and all others is mandatory.
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Priority Screens & Ownership

/auth/login Intern 4 (UI) + Intern 1 (API)	/superadmin/dashboard Intern 2	/superadmin/schools Intern 2	/superadmin/courses/builder Intern 2
/school-admin/dashboard Intern 3	/school-admin/attendance Intern 3	/student/dashboard Intern 4 ■	/student/course/[id] Intern 4
/student/lesson/[id] Intern 4	/student/achievements Intern 4	/parent/dashboard Intern 3 (Phase 3)	

Tech Stack Summary

Layer	Technology	Owner
Frontend Framework	Next.js 14 (App Router) + Tailwind CSS + shadcn/ui	All Interns
Animations / UX	Framer Motion · Lottie · Phaser.js (mini-games)	Intern 4
State Management	Zustand + React Query (TanStack Query)	All Interns
Backend Runtime	Node.js with Next.js API Routes	Intern 1
Database	PostgreSQL via Prisma ORM	Intern 1
Authentication	NextAuth.js + JWT with refresh token rotation	Intern 1
File / Video Storage	AWS S3 or Cloudflare R2 — signed expiring URLs	Intern 2
Video Streaming	HLS via FFmpeg + S3, or Bunny.net / Mux	Intern 2
AI Features	OpenAI GPT-4o — quiz & activity generation	Intern 2
Email / Alerts	Nodemailer + Resend for email; SMS for attendance	Intern 3
PDF Generation	react-pdf / Puppeteer — receipts, reports, certs	Intern 3

Layer	Technology	Owner
Deployment	Vercel (frontend) + Railway/Render (backend/DB)	Intern 1
Payment (Phase 3)	Razorpay — India-friendly payment gateway	Intern 1

Gamification Mechanics (Intern 4 Spec)

Element	Rule / Detail
XP Points	Complete video: +30 XP Pass quiz (first try): +50 XP Perfect score: +80 XP Daily login: +10 XP
Levels	Seedling (0–200) → Explorer (200–500) → Adventurer (500–1000) → Champion (1000+)
Streak	Login + complete ≥1 lesson daily = streak. 3-day bonus: 2× XP. 7-day: Special badge
Badges	20+ badges: First Lesson, Quiz Master, Speed Learner, Perfect Week, Top 3 in Class, etc.
Leaderboard	Class-level only (friendly competition). Resets monthly.
World Map	Each module = an island. Each lesson = a level. Completed = glowing, locked = foggy
Coins	Earn coins → spend in Avatar Shop (unlock avatars, backgrounds, mascot outfits)

Recommended Folder Structure

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/project-root
■ ■ ■ /frontend (Next.js)
■ ■ ■ ■ /app
■ ■ ■ ■ /superadmin ← Intern 2
■ ■ ■ ■ /school-admin ← Intern 3
■ ■ ■ ■ /student ← Intern 4
■ ■ ■ ■ /parent ← Intern 3 (Phase 3)
■ ■ ■ ■ /auth ← Intern 1 + 4
■ ■ ■ ■ /components
■ ■ ■ ■ /ui (shadcn base – All)
■ ■ ■ ■ /admin (Admin components – Intern 2 & 3)
■ ■ ■ ■ /student (Gamified components – Intern 4)
■ ■ ■ ■ ■ MascotGuide.tsx
■ ■ ■ ■ ■ XPBar.tsx
■ ■ ■ ■ ■ VideoPlayer.tsx
■ ■ ■ ■ ■ QuizEngine.tsx
■ ■ ■ ■ ■ MiniGame.tsx
■ ■ ■ ■ ■ WorldMap.tsx
■ ■ ■ ■ /shared
■ ■ ■ ■ /lib
■ ■ ■ ■ auth.ts ← Intern 1
■ ■ ■ ■ db.ts ← Intern 1 (Prisma client)
■ ■ ■ ■ utils.ts
■
■ ■ ■ /backend (if separate Express server)
■ ■ ■ ■ /routes ← Intern 1
■ ■ ■ ■ /controllers ← Intern 1
■ ■ ■ ■ /middleware ← Intern 1
■ ■ ■ ■ /services ← Intern 1
■ ■ ■ ■ /prisma ← Intern 1 (schema + migrations)
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■ ■ ■ /shared (types, constants – All interns)

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